

Javascript

Strings and it's methods

Theory

1. Strings in JavaScript

Definition: Strings are sequences of characters used to represent text. **Key point:** Strings are immutable — once created, they cannot be changed, but new strings can be formed from operations.

Syntax:

```
let str = "Hello World";
```

2. Common String Methods

- **Length** → `str.length` returns the number of characters.
- **Character access** → `charAt(index)` returns character, `charCodeAt(index)` returns Unicode.
- **Search** → `indexOf()`, `lastIndexOf()`, `includes()`, `startsWith()`, `endsWith()`.
- **Substring extraction** → `substring(start, end)`, `slice(start, end)` (supports negative indices).
- **Replace** → `replace(old, new)` replaces substring.
- **Split** → `split(separator)` converts string into array.
- **Case conversion** → `toUpperCase()`, `toLowerCase()`.
- **Trim** → `trim()` removes whitespace at start and end.

3. Template Literals

Definition: Introduced in ES6, template literals allow embedding expressions inside strings using backticks.

Syntax:

```
let name = "Hema";
```

```
let age = 20;
```

```
let word = `${name} is ${age} years old`;
```

4. String Conversion

- `String(value)` → converts numbers, booleans, or other types into strings.

Example:

```
String(100); // "100"
```

```
String(true); // "true"
```

Coding Round Questions

- String length → Write a program to find the length of a string.
- Substring vs Slice → Demonstrate difference between `substring()` and `slice()`.
- Replace → Replace all occurrences of a word in a string.
- Split → Split a sentence into words using `split()`.
- Template literals → Use template literals to print a sentence with variables.
- Trim → Show how `trim()` removes whitespace.

Machine Coding Round Challenge

Problem: User Registration System

Requirements:

- Accept user input: name, email, password.
- Validate:
 - Name should not be empty.
 - Email should contain @ and end with .com.
 - Password should be at least 8 characters.
- Use string methods (`includes`, `endsWith`, `length`, `trim`) for validation.
- Print success or error messages.

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Strings</title>
  </head>
  <body>
    <h2>Strings</h2>
    <script>
      let str = "Hello World fiit academy";

      console.log(str.length);

      console.log(str.charAt(7));

      console.log(str.charCodeAt(0));

      console.log(str.indexOf("o"));

      console.log(str.lastIndexOf("o"));

      console.log(str.substring(0, 5));

      console.log(str.slice(-5)); //supports negative index

      console.log(str.replace("H", "A"));

      console.log(str.split(" ")); // important

      console.log(str.toUpperCase());

      console.log(str.toLowerCase());

      console.log(str.trim()); // trims space in start and end

      console.log(str.startsWith("dello"));

      console.log(str.endsWith("y"));

      console.log(str.includes("e"));

      let name = "Hema";
      let age = 20;

      let word = `${name} is ${age} years old`;

      console.log(word);
    </script>
  </body>
</html>
```

```
console.log([...str]);

// number to string

let num = String(100);

console.log(typeof num);

console.log(String(true));

</script>
</body>
</html>
```